

Why we filter COCONUT

the case for coconut-flavordb-filtered, and what it costs

The decision

MolPLAtte pretrains on molecular decompositions: a core plus its detached R-groups. The R-group vocabulary that emerges is set by whatever chemistry dominates the corpus. We combine two sources, and they are wildly unequal in both size and relevance:

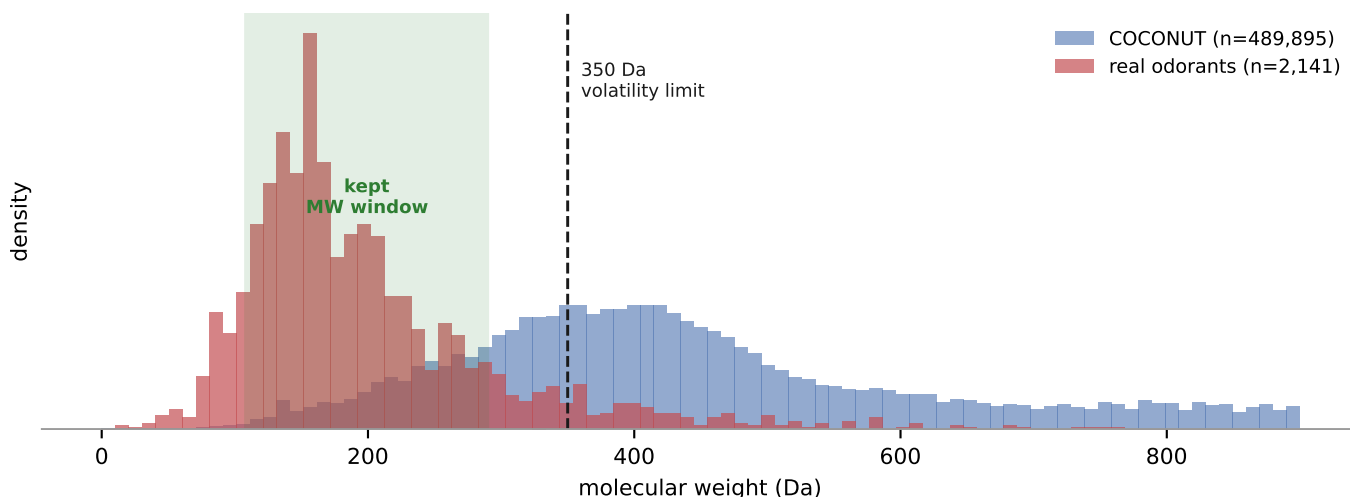
FlavorDB	20,812 records	every molecule has a measured flavour label
COCONUT	374,454 records	natural products; 0.66% have any known flavour

Unfiltered, COCONUT outnumbers FlavorDB 18 to 1, so the vocabulary is learned almost entirely from natural-product chemistry — macrolides, glycosides, polysaccharides — rather than from flavour chemistry. `coconut-flavordb-filtered` restricts COCONUT to the physicochemical envelope real odorants occupy, buying scale without handing the vocabulary to the wrong domain.

The size mismatch is not subtle

COCONUT's median molecular weight is 427 Da, and 69.9% of it sits above 350 Da. Real odorants — FlavorDB entries with a measured, non-sweet sensory label — have a median of 170 Da, with only 7.9% above 350.

That 350 Da line is not arbitrary. A molecule must volatilise to reach an olfactory receptor, and above roughly 350 Da it effectively cannot. Most of COCONUT is therefore not merely unlabelled — it is physically incapable of having an odour.



The distributions barely overlap. Training on their union unfiltered means R-group statistics are set by the blue distribution while the task we care about lives in the red.

A caution about what counts as an odorant

FlavorDB's `flavor\_profile` column is populated for 25,106 of its 25,596 rows, so "has a label" is nearly vacuous — that set has a median MW of 279 Da, describing FlavorDB's bulk rather than its odorants. The envelope below is derived only from the 2,141 entries whose profile maps onto a real sensory class and is not purely sweet. Sweet compounds are excluded because taste receptors sit in solution, so sweetness carries no volatility constraint — steviosides are intensely sweet at roughly 800 Da.

The envelope, and what it buys

derived from measured odorants — never hand-picked

A COCONUT compound is kept if it falls inside all three bounds:

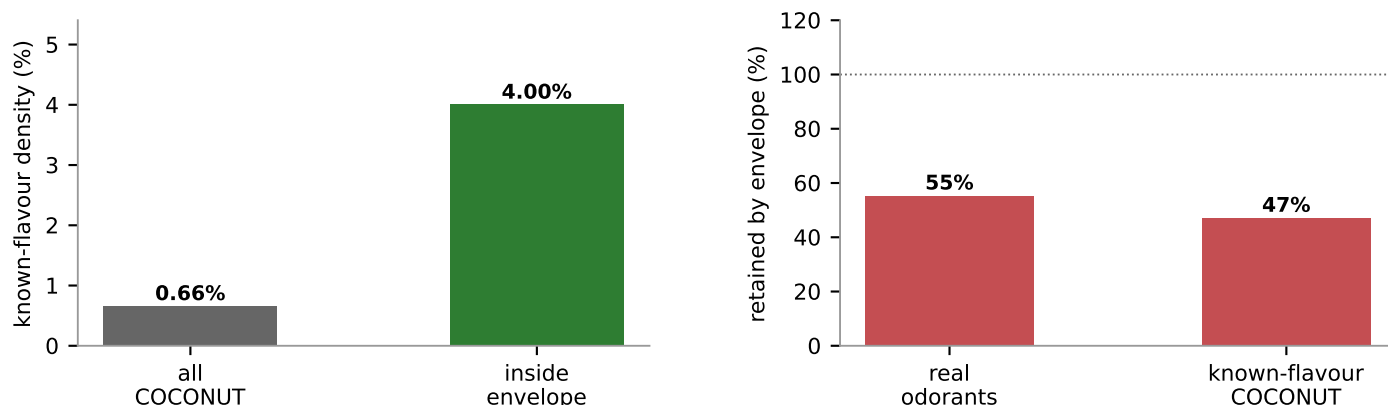
molecular weight	108 – 290 Da
logP	0.4 – 4.5
topological PSA	≤ 53

These are percentiles of the 2,141 measured odorants, not intuition. The filter keeps 37,776 of 489,895 COCONUT compounds — 7.7%.

Enrichment: does it concentrate flavour chemistry?

The honest test is whether compounds independently known to be flavour-relevant (present in FlavorDB by InChIKey) are denser inside the envelope than outside.

base rate across COCONUT	0.66%	(3,224 of 489,895)
inside the envelope	4.00%	(1,512 of 37,776)
enrichment	6.1x	



Left: the filter concentrates flavour chemistry 6.1-fold. Right: it is also blunt — it discards 45% of real odorants and 53% of COCONUT compounds already known to carry flavour.

Why the corpus is a union, not just the envelope

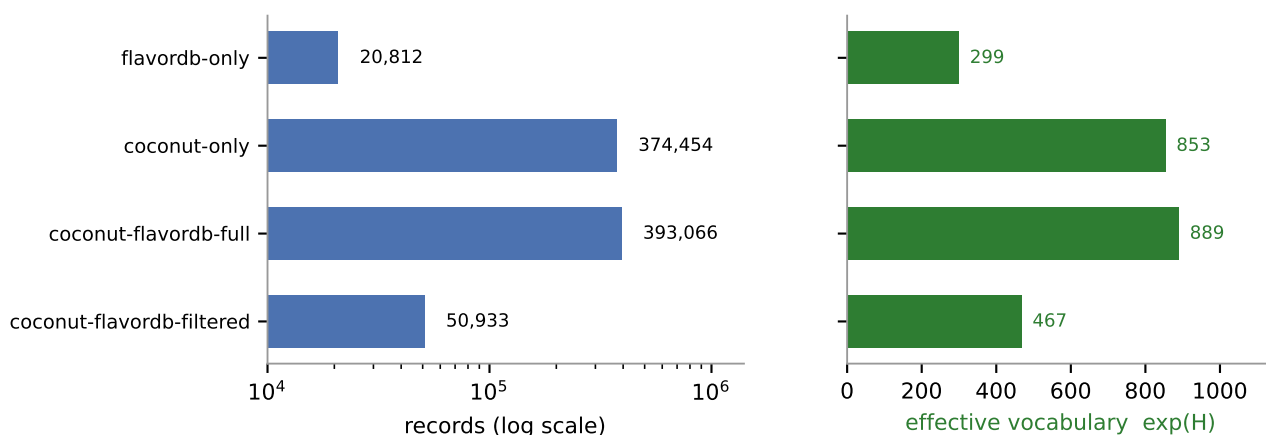
A 47% recall on known-flavour compounds is unacceptable: those are the most valuable molecules in the corpus, and a three-descriptor box discards more than half of them. The built corpus therefore keeps the envelope UNION every compound carrying a known flavour label. That restores recall on the known set to 100% and yields an allowlist of 39,401 COCONUT identifiers.

One consequence deserves stating plainly, because it is easy to quote misleadingly. The known-flavour density of that union is 5.98%, which against the 0.66% base rate reads as a 9x enrichment. That figure is partly circular: the union was built by adding every known positive, so its density is inflated by construction. The defensible number for what the FILTER itself achieves is the envelope-only 6.1x above.

What filtering costs

and when to reach for which corpus

Filtering is a trade, not a free improvement: it buys domain density and pays in scale and vocabulary breadth. All four corpora share build tag `r333-m2-h4-flavor24`, so they are directly comparable.



Against `coconut-flavordb-full`, the filtered corpus is 7.7x smaller and its effective vocabulary — $\exp(\text{entropy of the R-group distribution})$, i.e. the number of R-groups actually carrying probability mass — falls from 889 to 467, roughly half.

This matters for evaluation. Retrieval difficulty scales with effective vocabulary, so Hit@K on the filtered corpus is NOT comparable to Hit@K on the full one. The filtered task is easier by construction, and a higher number there is not evidence of a better model.

Which corpus for which question

coconut-flavordb-full default for pretraining: maximum scale and the broadest R-group vocabulary, with sparse flavour labels.

coconut-flavordb-filtered pocket finetuning and flavour conditioning, where label density matters more than raw scale.

coconut-only the contrast arm: natural-product chemistry with almost no flavour labels. Any claimed flavour effect should be checked against this to confirm it does not reproduce.

flavordb-only small and fully labelled; use when label coverage is the binding constraint.

Open limitations

The envelope is a three-descriptor axis-aligned box. Real odorancy is not box-shaped, and the 45% of true odorants it rejects is the price of that simplicity. The union with known-flavour compounds patches the recall we can measure; it cannot patch recall on flavour-relevant compounds nobody has labelled yet.

Enrichment is measured against FlavorDB membership, itself a biased sample of flavour chemistry that over-represents well-studied food compounds. Read the 6.1x as enrichment in DOCUMENTED flavour relevance, not in flavour relevance as such.